

PHITCHANAN SAWATAUE

Architect

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EDUCATION

Doctor of Philosophy (Ph.D.) in Engineering

Present

National Taiwan University of Science and Technology (NTUST)

- Focus: Computational Intelligence, Building Information Modeling (BIM), and Digital Twin technology.

Master of Architecture

Dec 2017 - Jul 2019

Chulalongkorn University

- Major in Architecture, specializing in Architectural Management
- Thesis: "The Study of Changes in the Scope of Construction Supervision Duties in Government Projects"

Bachelor of Architecture

Jun 2008 - Mar 2013

King Mongkut's Institute of Technology Ladkrabang

- Major in Architecture
- Thesis: "Architectural Design for the Institute of Arabica Coffee Research and Development"

Vocational Certificate in Architectural Technology

May 2005 - Mar 2008

Asia Technological College

- Relevant coursework: Architectural Technical Studies

PROFESSIONAL AFFILIATIONS

- Registered Architect, Council of Architects Thailand (Professional Architect License Arch. 3747)
- Member, Association of Siamese Architects (ASA)
- Member, Thai BIM Association

WORK EXPERIENCE

Architect | Terra Co., Ltd.

Jul 2018 - Present

- Lead architect for various residential, commercial, and industrial projects.
- Specialized in Building Information Modeling (BIM) and architectural planning.
- Collaborated with legal and environmental research teams to ensure compliance with building regulations.
- Assisted in architectural research projects at Chulalongkorn University's Building and Environmental Law Research Unit.

Freelance Architect & Research Assistant

2017 - 2018

- Designed architectural projects for private clients.
- Supported research at Chulalongkorn University's Academic Service Center, focusing on building laws and environmental regulations.

- Designed and developed residential, public, and industrial buildings.
- Coordinated with engineers and contractors for project execution and construction management.
- Conducted feasibility studies and prepared technical drawings for large-scale projects.

ACADEMIC & RESEARCH ROLES

Research Assistant | Chulalongkorn University

2018 - Present

- Contributed to the development of BIM-based facility management systems for over 201 university buildings.
- Led heritage building analysis projects using advanced Point Cloud and Photogrammetry techniques, contributing to conservation efforts and regulatory compliance.
- Developed strategies for integrating BIM with regulatory compliance, sustainability, and safety inspections.
- Assisted in the Thai Red Cross Society's building safety audits under national compliance laws.

Part-time Lecturer | Chulalongkorn University

2019 - Present

- Taught courses in Architectural Design, Building Technology, and Construction Materials across multiple academic years.
- Instructed Architectural Presentation and Construction Materials for first-year students.
- Developed course content focusing on digital modeling, BIM applications, and sustainable design.
- **Courses Taught:** Architectural Design and Integrated Building Technology (2019-2024, Year 5), Architectural Materials and Construction I (2022, Year 1), Architectural Presentation I (2019, Year 1).

Teaching Assistant (TA) | Chulalongkorn University

2017 - 2019

- Assisted in teaching and curriculum development for courses in Building Technology, Construction Management, and Environmental Law.
- Provided academic support and guidance to third-year students in Architectural Design Studio.
- Conducted tutorials on Computer-Aided Design and Innovation for architecture students.
- **Courses Assisted:** Building Technology and Construction IV, Building and Environmental Law, Construction Management, Computer-Aided Architectural Design Innovation, Architectural Design Studio V (Year 3).

RESEARCH INTERESTS

- **BIM & AI Optimization:** Utilizing Metaheuristic Algorithms for smart construction and resource-efficient planning.
- **Heritage BIM:** Applying Point Cloud & Photogrammetry for digital conservation of historical buildings.
- **GIS & BIM Integration:** Enhancing urban planning and facility management through geospatial data analytics.
- **Sustainable Building Design:** Investigating passive cooling strategies and parametric design for energy-efficient architecture.

ADDITIONAL INFORMATION

Technical Skills:

AutoCAD (Advanced) Revit (Proficient) SketchUp (Proficient) BIM (Expert)

Languages: Thai, English